

Data Analytics

What is Analytics?

Analytics is the **systematic process of collecting, processing, and analyzing data** to uncover meaningful insights, patterns, and trends that can help in **decision-making**.

In simple terms, **analytics = data + analysis → insights + action**.

Key Idea

Analytics turns *raw data* into *useful information*.

For example:

- A company uses **sales analytics** to see which products sell best.
- A website uses **web analytics** to understand visitor behavior.
- A hospital uses **healthcare analytics** to improve patient care.

Types of Analytics

1.Descriptive Analytics – *What happened?*

→ Example: “Sales increased by 20% last quarter.”

2.Diagnostic Analytics – *Why did it happen?*

→ Example: “Sales increased because of a holiday promotion.”

3.Predictive Analytics – *What might happen next?*

→ Example: “Sales are expected to rise 15% next quarter.”

4.Prescriptive Analytics – *What should we do?*

→ Example: “Increase marketing budget in high-performing regions.”

What is Predictive Analytics?

Predictive Analytics is a branch of data analytics that focuses on **forecasting future outcomes** using **historical data**, **statistical algorithms**, and **machine learning techniques**.

In simple terms:

Predictive Analytics = Using past data to predict future events.

Example

- A **bank** uses predictive analytics to identify which customers are likely to default on loans.
- An **e-commerce site** predicts what products a customer might buy next.
- A **hospital** forecasts which patients are at high risk of readmission.

How Predictive Analytics Works

- 1. Collect Data** – Gather historical and current data (e.g., sales, customer behavior, weather, etc.).
- 2. Clean & Prepare Data** – Remove errors, fill missing values, and format the data.
- 3. Build Models** – Use techniques like regression, decision trees, or neural networks to find patterns.
- 4. Test & Validate** – Check how accurate the model's predictions are.
- 5. Deploy & Monitor** – Use the model to make real-world predictions and improve it over time.

Common Techniques

- **Regression Analysis** – Predicts a continuous value (e.g., sales next month).
- **Classification Models** – Predicts categories (e.g., "will churn" or "won't churn").
- **Time Series Analysis** – Forecasts data over time (e.g., stock prices, demand trends).
- **Machine Learning** – Uses algorithms that learn patterns automatically from data.

Discrete Vs. Continuous Data

1. Discrete Data

Data that can take **specific, separate values** — usually **countable** numbers.

You **cannot have fractions or decimals** between these values.

Examples:

- Number of students in a class → 30 (you can't have 30.5 students)
- Number of cars in a parking lot → 12
- Number of goals scored in a match → 3

Key points:

- Comes from **counting**
- Has **finite or countable** possible values
- Represented by **whole numbers**

2. Continuous Data

Data that can take **any value within a range** — often **measurable** rather than countable.

You **can have decimals or fractions**.

Examples:

- Height of a person → 172.5 cm
- Weight of a baby → 3.27 kg
- Temperature → 25.3°C

Key points:

- Comes from **measuring**
- Can take **infinite possible values** within a range
- Includes **decimals and fractions**

Supervised Vs. Unsupervised Learning

1. Supervised Learning

Supervised learning is when the model is **trained using labeled data** — meaning we already know the **input (X)** and the **correct output (Y)**.

The algorithm learns the relationship between them so it can predict outcomes for new data.

In simple terms:

You teach the model with examples that already have answers.

Examples:

- **Email classification:** Spam or Not Spam
- **House price prediction:** Predict price based on size, location, etc.
- **Disease diagnosis:** Predict if a patient has diabetes (Yes/No)

Common Algorithms:

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machine (SVM)
- Neural Networks

2. Unsupervised Learning

Unsupervised learning is when the model is **trained on unlabeled data** — meaning we only have inputs (**X**) and **no predefined outputs (Y)**.

The model tries to **find hidden patterns or structure** in the data on its own.

In simple terms:

The model learns **without answers**, discovering patterns by itself.

Examples:

- **Customer segmentation:** Group customers by buying behavior.
- **Music recommendation:** Cluster songs based on listening patterns.
- **Anomaly detection:** Detect unusual transactions or fraud.

Common Algorithms:

- K-Means Clustering
- Hierarchical Clustering
- Principal Component Analysis (PCA)

Parametric Vs. Non-Parametric Models

Parametric Models

These models **make strong assumptions** about the **underlying data distribution** and have a **fixed number of parameters**.

Once those parameters are learned, the model is fully defined.

In simple terms:

A parametric model assumes a specific *shape/distribution* for the data (like a straight line or a normal curve).

Examples:

- Linear Regression
- Logistic Regression
- Naïve Bayes
- Linear Discriminant Analysis (LDA)

Characteristics:

- Fixed number of parameters (doesn't grow with data)
- Assumes a specific functional form (e.g., linear, Gaussian)
- Fast and easy to interpret
- May perform poorly if the assumption is wrong

Example (Linear Regression): $y = mx + b$

Only two parameters (slope m , intercept b) describe the model — no matter how many data points you have.

Non-Parametric Models

These models **do not assume** a specific form for the data distribution and can have a **flexible, data-driven structure**.

The number of parameters **can grow with the amount of data**.

In simple terms:

A non-parametric model lets the data decide its own shape — it's more flexible.

Examples:

- K-Nearest Neighbors (KNN)
- Decision Trees / Random Forests
- Support Vector Machines (with non-linear kernels)
- Neural Networks (often considered semi-parametric)

Characteristics:

- Flexible — can fit complex patterns
- Fewer assumptions about the data
- Require more data and computational power
- Can overfit if not regularized

Predictive Analytics Vs. Business Intelligence

Business Intelligence (BI)

To provide **descriptive insights** — “What happened?” and “Why did it happen?”

Fraud Detection:

- How many cases were investigated last month?
- How much revenue was recovered through collections?

Customer Analytics:

- What region had the highest response rate?
- How many repeat purchasers were there in last month?
- How many visit to the store did a person have?

Predictive Analytics

To provide **predictive insights** — “What will happen?” or “What is likely to happen?”

Fraud Detection:

- What is the likelihood that the transaction is fraudulent?
- What is the expected mount of fraud?

Customer Analytics:

- What product is a customer most likely to purchase if given the choice?
- What is the likelihood a product will sell out if it is put on sale?

Predictive Analytics Vs. Statistics

Statistics

It focuses on **understanding relationships** and **drawing conclusions** about data — often to test hypotheses or describe populations. **Theory-driven approach.**

Statistics helps you **understand what your data says** and whether patterns are *real or random*.

Focus on: Probabilities, significance tests, confidence intervals

Goal:

- Explain or describe what happened
- Test relationships and significance (e.g., correlation, cause–effect)

Example Questions:

- “Is there a relationship between education and income?”
- “Is the difference between two groups statistically significant?”

Predictive Analytics

It's more about **making accurate predictions** than explaining why things happen. **Data-driven approach.**

Predictive analytics helps you **use data to guess the future.**

Focus on: Forecasts, risk scores, likelihoods

Goal:

- Forecast what will happen next
- Support data-driven decision-making

Example Questions:

- “Which customers are likely to cancel their subscription?”
- “What will next month's sales be?”

Predictive Analytics Vs. Data Mining

Data Mining

Data mining is the **process of discovering patterns, correlations, and relationships** in large datasets — **without necessarily predicting** anything.

It's about **finding hidden patterns and knowledge** in data.

Example Questions:

- “What patterns exist in customer purchase behavior?”
- “Which products are often bought together?”
- “Are there natural groupings in my data?”

Typical Techniques:

- Clustering (e.g., K-Means)
- Association Rules (e.g., Market Basket Analysis)
- Anomaly Detection
- Summarization and Pattern Recognition

Challenges in using Predictive Analytics

1. Data-Related Challenges

I. Poor Data Quality

- Incomplete, inconsistent, or inaccurate data can lead to wrong predictions.
- Example: Missing sales records or incorrect customer info.

II. Insufficient Data

- Predictive models require **large amounts of historical data** to learn patterns.
- Small datasets can result in overfitting or unreliable predictions.

III. Data Silos

- Data stored in different departments or systems may not be integrated, making analysis difficult.

2. Model-Related Challenges

I. Choosing the Right Model

- Different algorithms suit different problems; picking the wrong one can reduce accuracy.

II. Overfitting or Underfitting

- **Overfitting:** Model learns noise instead of patterns → poor on new data.
- **Underfitting:** Model is too simple → misses important patterns.

III. Model Interpretability

- Complex models like deep neural networks are **hard to interpret**, making it difficult to explain predictions to stakeholders.

3. Operational Challenges

I. Integration into Business Processes

- Predictions are only useful if they are **actionable** and integrated into decision-making workflows.

II. Scalability & Performance

- Handling large datasets in real-time or near real-time can be computationally expensive.

III. Keeping Models Updated

- Models trained on old data can become outdated as business conditions or customer behavior

4. Human & Organizational Challenges

I. Lack of Skilled Personnel

- Predictive analytics requires expertise in statistics, machine learning, and domain knowledge.

II. Resistance to Change

- Decision-makers may distrust predictions and prefer traditional methods.

III. Ethical & Privacy Concerns

- Using personal or sensitive data for predictions raises privacy issues and regulatory compliance challenges (e.g., **General Data Protection Regulation (GDPR)**).

5. External Challenges

I. Unpredictable Events

- Models can fail to predict rare or unprecedented events (e.g., natural disasters, pandemics).

II. Bias in Data

- If historical data contains biases, predictions can **perpetuate unfair or discriminatory outcomes**.

Data Understanding: What the data looks like?

It's all about **knowing your data before analyzing or modeling it.**

1. Collect Initial Data

- Gather all relevant datasets from different sources (databases, CSV files, APIs, etc.).
- Understand **where the data comes from** and its format.
- Example: Collect sales data, customer info, and website activity logs.

2. Describe the Data

- Look at **basic statistics** and properties of the data.
- Examples:
 - Number of rows and columns
 - Data types (numeric, categorical, date/time)
 - Min, max, mean, median values
 - Frequency distributions for categorical variables

3. Explore the Data (Exploratory Data Analysis)

- Identify **patterns, trends, and anomalies**.
- Look for **outliers, missing values, and inconsistencies**.
- Visualize data using histograms, scatter plots, box plots, heatmaps.

4. Verify Data Quality

- Check for **missing values, duplicates, inconsistent formats, and errors**.
- Assess **completeness, accuracy, and reliability** of the data.
- Example: Customer age should not be negative; email addresses should have valid formats.

5. Understand Relationships

- Look at **correlations and interactions** between variables.
- Example: Sales may correlate with advertising spend or seasonality.
- Helps identify which variables might be **useful for predictive modeling**.

6. Document Insights and Issues

- Keep notes of **key findings, anomalies, and assumptions**.
- Document any **limitations or biases** in the dataset.
- Example: “Data from 2020 has missing entries for Q3 due to system outage.”

Single Variable Summaries

A. For Numerical (Quantitative) Data

These are used for variables like age, salary, height, etc.

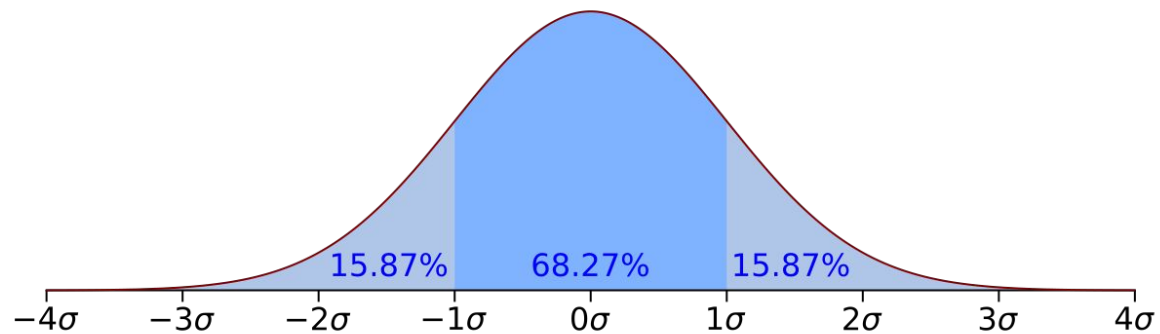
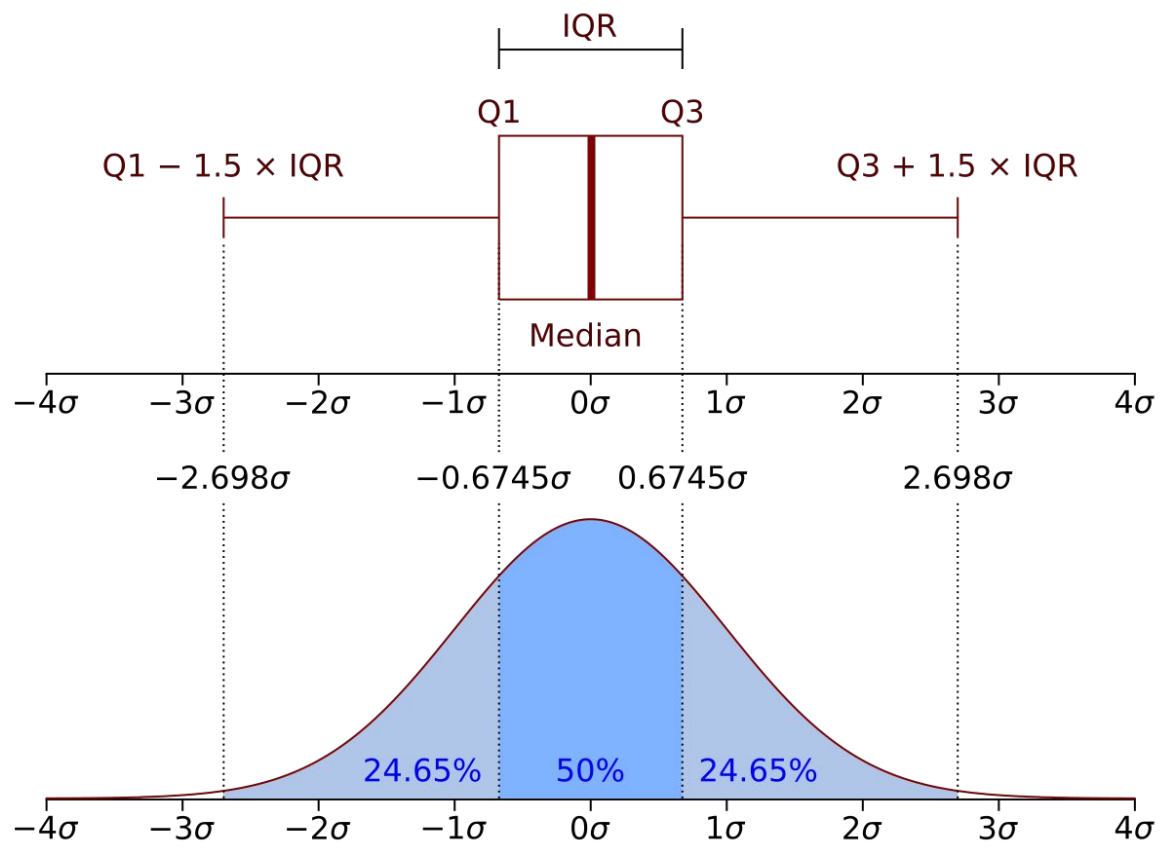
Common Summaries:

1. Measures of Central Tendency – Describe the “center” of the data:

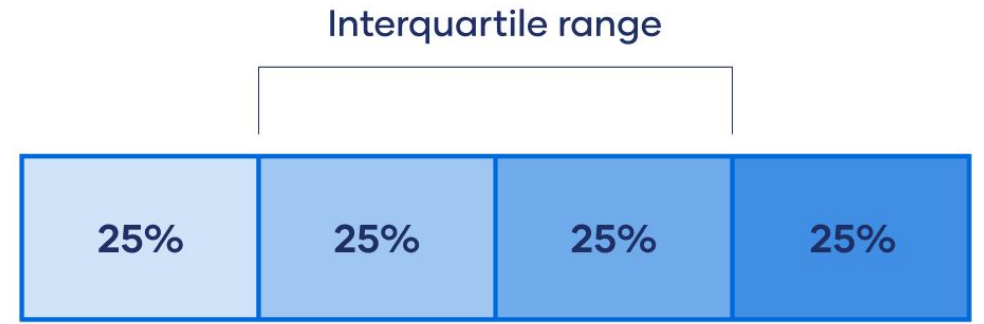
1. **Mean:** Average value
2. **Median:** Middle value when data is sorted
3. **Mode:** Most frequent value

2. Measures of Dispersion – Describe how spread out the data is:

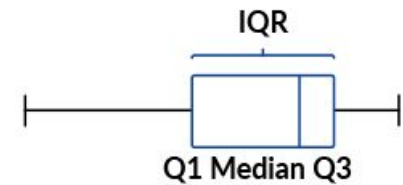
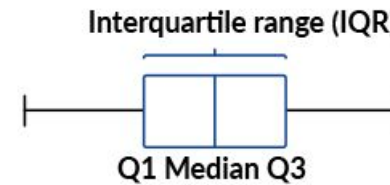
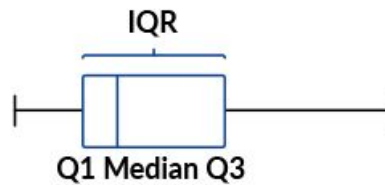
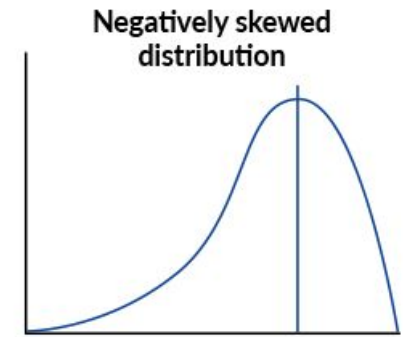
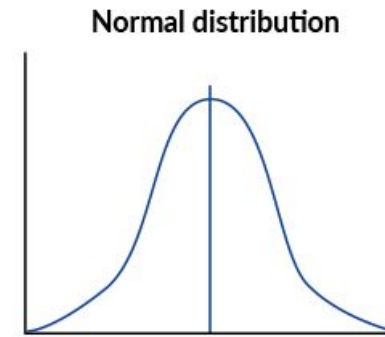
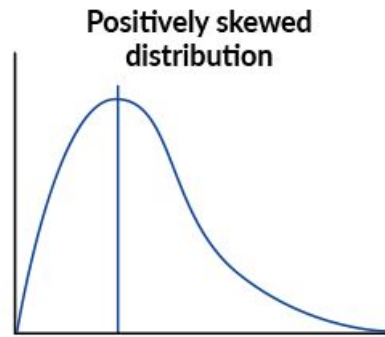
1. **Range:** Max – Min
2. **Variance:** Average squared deviation from the mean
3. **Standard Deviation (SD):** Square root of variance
4. **Interquartile Range (IQR):** Difference between Q3 and Q1



probability density function (pdf) of a Normal $N(0, \sigma^2)$ Population



Q1 Q2 Q3



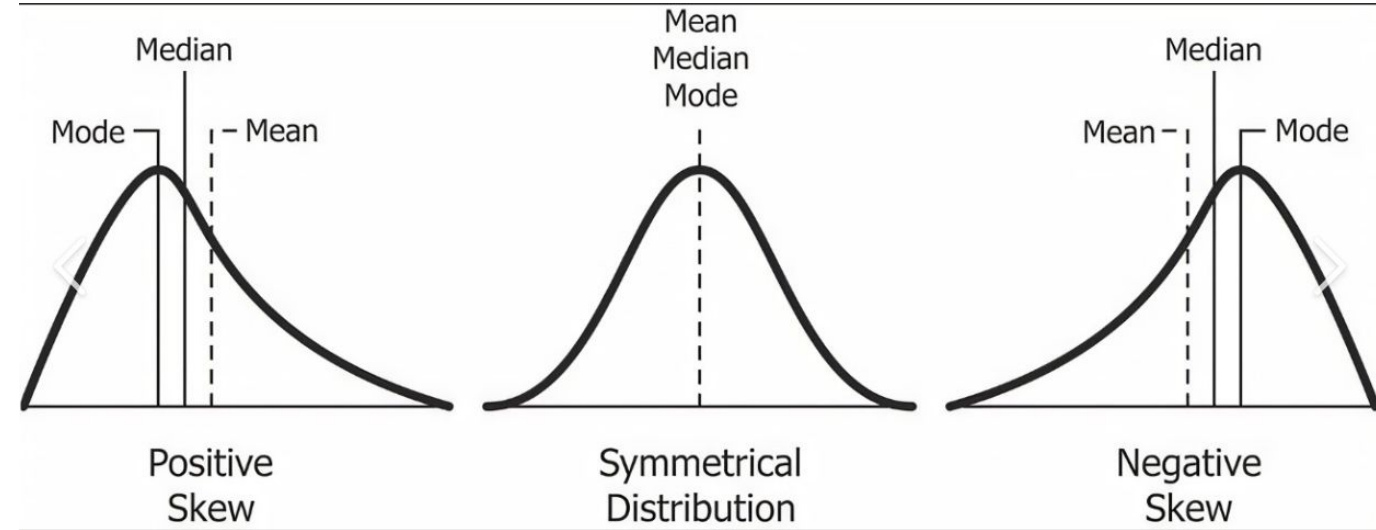
- The IQR tells how **spread out** the middle half of the data is.
- The IQR can be used to **identify outliers**.
- The IQR also may **indicate the skewness** of the dataset.
- It is **not affected by outliers** unlike the range

3. Shape of Distribution – Understand the pattern:

1. **Skewness:** Asymmetry of the data
2. **Kurtosis:** How “peaked” the distribution is?

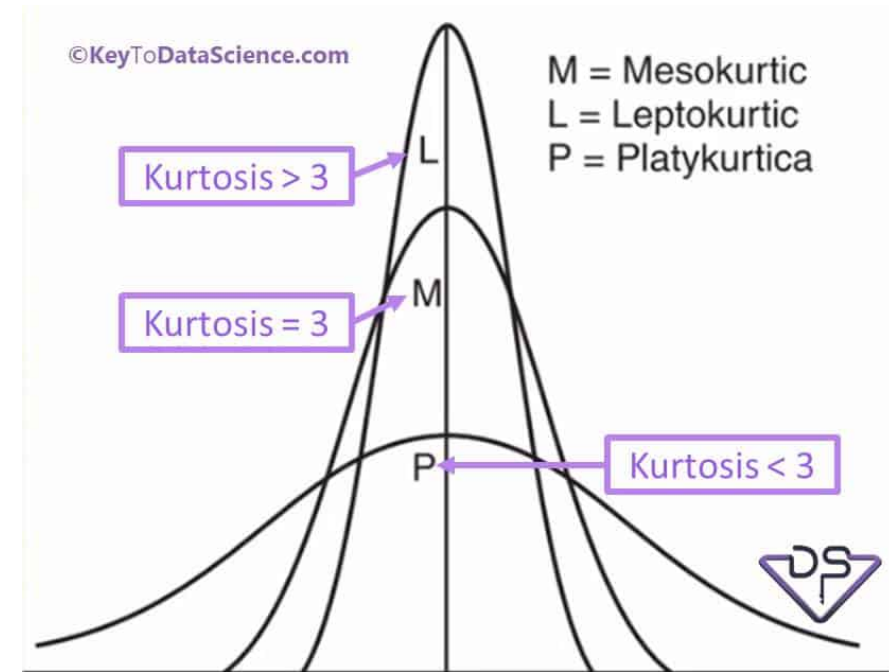
Or

How much thinner or fatter the distribution is compared to the normal distribution?



4. Visualizations:

1. Histogram
2. Box plot
3. Density plot



B. For Categorical (Qualitative) Data

These are used for variables like gender, product type, or country.

Common Summaries:

1.Frequency Counts:

- How many times each category appears

2.Proportions / Percentages:

- Relative frequency of each category

3.Mode:

- Most common category

4.Visualizations:

- Bar chart
- Pie chart

Variable Type	Example	Summary Methods
Numerical	Age of employees	Mean = 35, Median = 34, SD = 5.2, Histogram shows right skew
Categorical	Department (HR, IT, Sales)	HR: 10, IT: 20, Sales: 15; Pie chart shows proportions

Why Single Variable Summaries Matter

- Helps understand distribution, central tendency, and variability.
- Identifies outliers or data entry errors.
- Forms the basis for further analysis, like relationships between variables.

Normal Distribution (Gaussian Distribution)

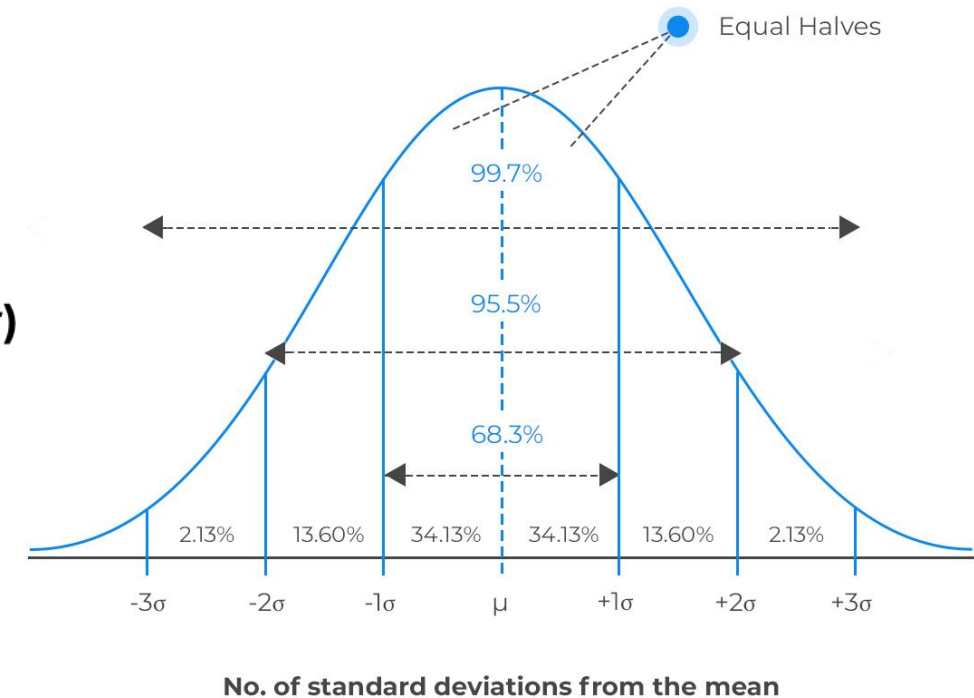
A normal distribution is a **continuous probability distribution** that is **symmetrical around the mean**, forming a **bell-shaped curve**. Most values cluster around the mean, and probabilities decrease as you move further away.

Key Properties:

- Symmetrical about the mean (μ)
- Mean = Median = Mode
- Defined by two parameters: **Mean (μ)** and **Standard Deviation (σ)**
- About 68% of data lies within 1 SD, 95% within 2 SD, and 99.7% within 3 SD (Empirical Rule)

Example:

- Heights of adult humans
- IQ scores
- Blood pressure readings



Uniform Distribution

A uniform distribution is a **continuous probability distribution** where **all values within a certain range are equally likely**. The graph is **flat**, showing no bias toward any value.

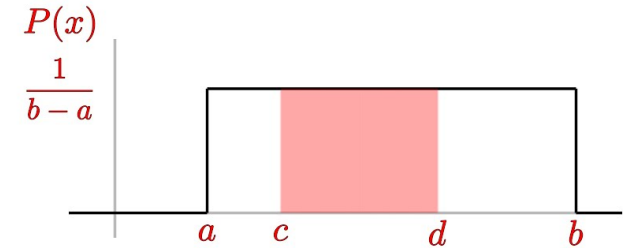
Key Properties:

- **Constant probability** across the range
- Mean = $(a + b)/2$, where a and b are the min and max values
- Variance = $(b - a)^2/12$

Example:

- Rolling a fair die (discrete uniform)
- Picking a random number between 0 and 1 (continuous uniform)
- Randomly selecting a card from a shuffled deck (discrete uniform)

Uniform Distribution



$$\text{Mean : } \mu = \frac{a + b}{2}$$

$$\text{S.D. : } \sigma = \sqrt{\frac{(b - a)^2}{12}} \quad \text{Probability } P(c \leq X \leq d) = \frac{d - c}{b - a}$$

1. Setup: Continuous Uniform Distribution

A continuous uniform distribution over the interval $[a, b]$ has:

- Probability density function (PDF):

$$f(x) = \frac{1}{b-a} \quad \text{for } a \leq x \leq b$$

- All values are **equally likely** between a and b .

Quick Memory Tip

- Normal → “Bell curve” → most values around the middle
- Uniform → “Flat line” → all values equally likely

2. Variance Formula

The **variance** of a continuous random variable X is:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Where:

- $E[X]$ = expected value (mean)
- $E[X^2]$ = expected value of X^2

Data Visualization in 1-Dimension

1-Dimensional Data Visualization refers to visualizing **a single variable** at a time to understand its **distribution, central tendency, spread, and outliers**.

- Useful in **Exploratory Data Analysis (EDA)**.
- Can be applied to **numerical** or **categorical** data.

1. Histogram

A **Histogram** is a **graphical representation of the distribution of numerical data**.

It **divides the data into intervals (bins)** and shows **how many data points fall into each bin**.

Key Features

1.Bins (Intervals):

1. Range of values divided into equal or meaningful segments.
2. Example: Ages 0–10, 11–20, etc.

2.Frequency / Count:

1. Height of each bar represents the number of data points in that bin.

3.Continuous Data Representation:

1. Bars **touch each other** (unlike bar charts for categorical data).

Purpose of a Histogram

- Understand **distribution of data** (shape, spread, central tendency).
- Identify **skewness** (left-skewed, right-skewed, symmetric).
- Detect **outliers** or unusual observations.
- Visualize **density patterns**.

Shape of Distribution

- **Symmetric (Normal-like):** Peak in the middle, tails on both sides.
- **Right-skewed (positive skew):** Tail on the right.
- **Left-skewed (negative skew):** Tail on the left.
- **Uniform:** Bars roughly equal height.

Example

Suppose exam scores of 15 students are:

45, 50, 52, 55, 57, 60, 63, 65, 67, 70, 72, 75, 78, 80, 85

Define **bins** (e.g., 40–50, 51–60, 61–70, 71–80, 81–90)

Count **frequencies**:

40–50 → 2 students

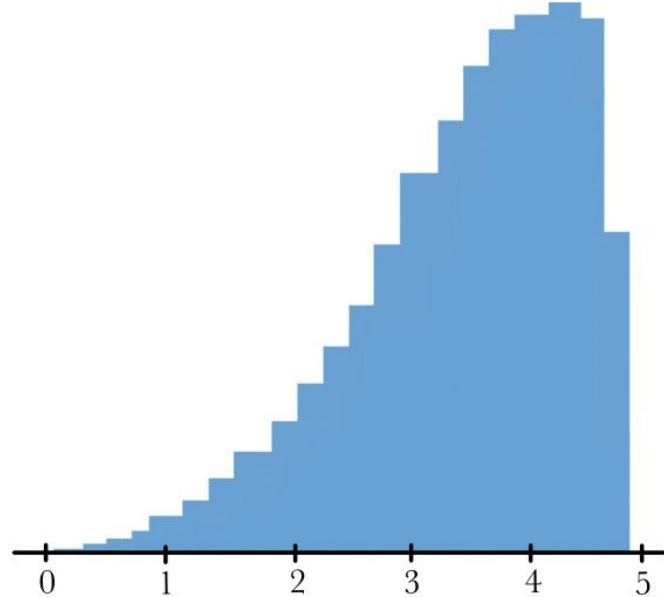
51–60 → 3 students

61–70 → 4 students

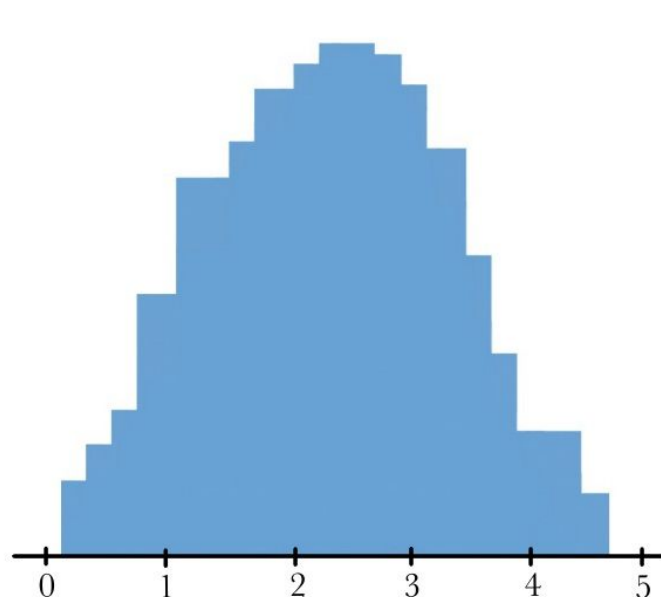
71–80 → 4 students

81–90 → 2 students

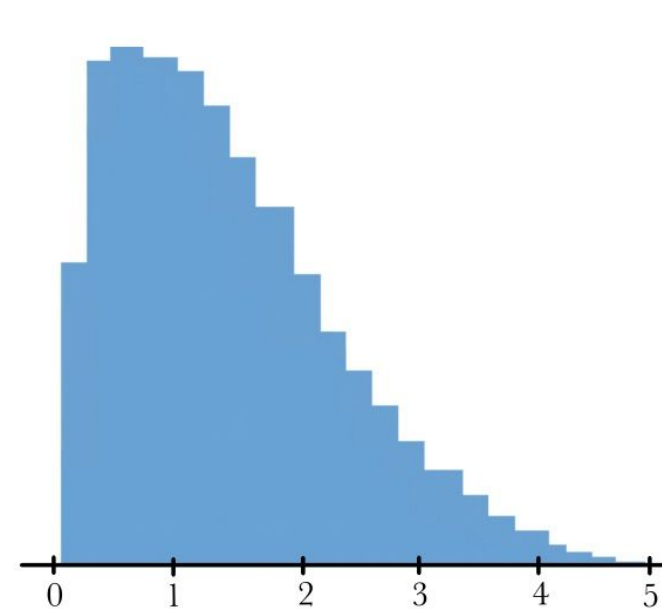
Draw bars **touching each other** with heights equal to frequencies.
This gives a visual sense of how scores are distributed.



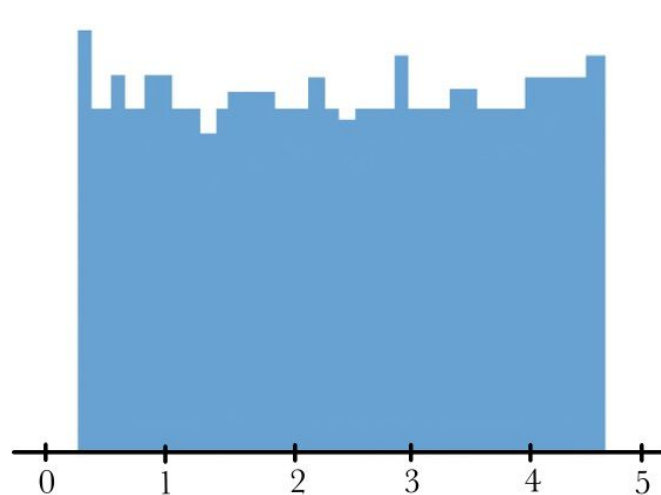
skew left



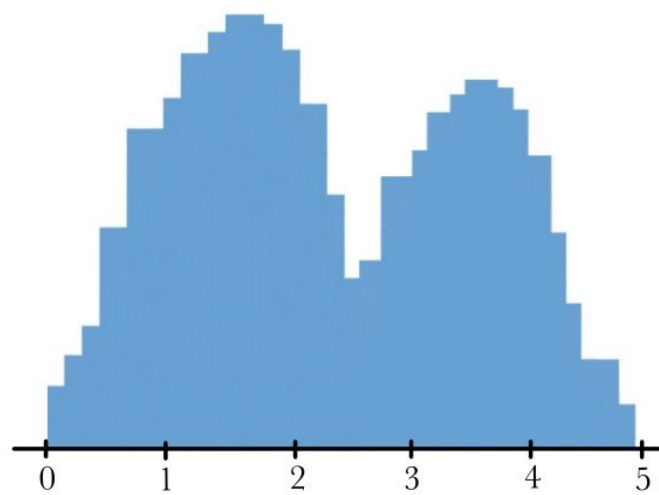
symmetric, unimodal



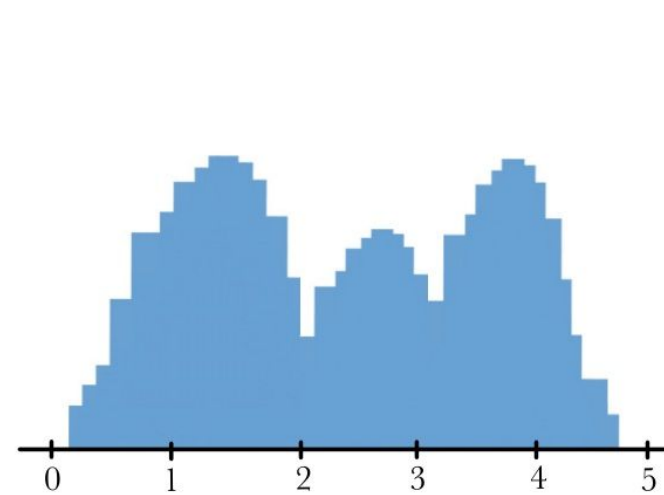
skew right



uniform



bimodal



multimodal

2. Box Plot

A **Box Plot** (also called a **Whisker Plot**) is a **graphical summary of a dataset** that shows the **distribution, central tendency, spread, and outliers**.

- It is especially useful for **comparing distributions** across different groups.

Components of a Box Plot

A box plot is made up of **five main statistics** (also called the **five-number summary**):

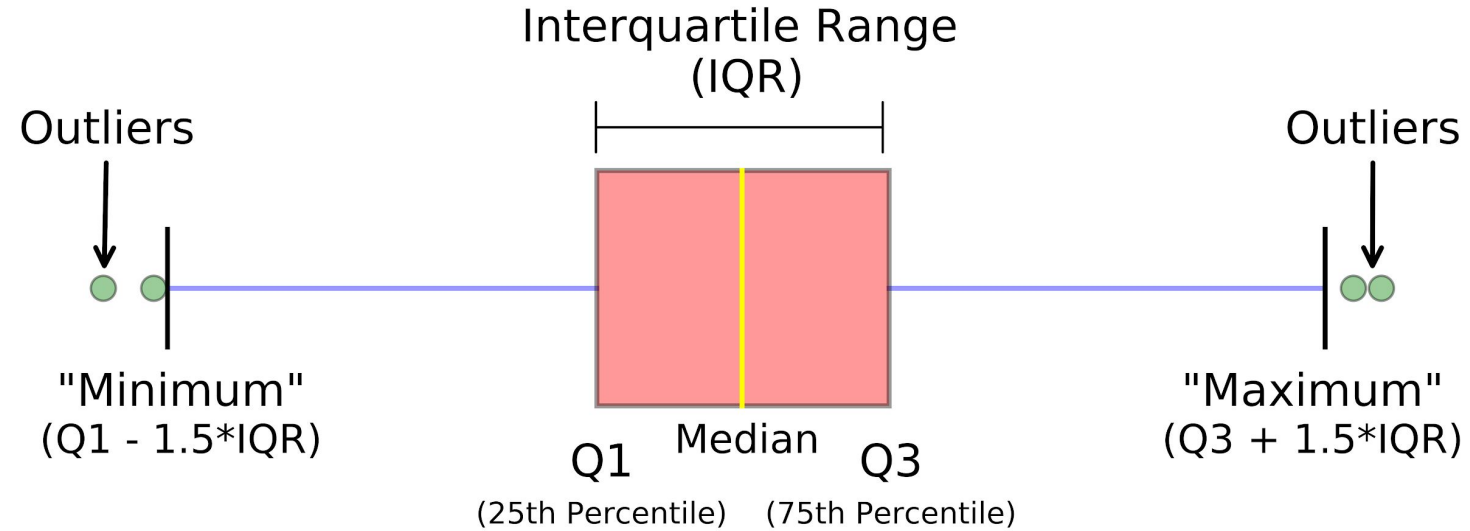
- 1.**Minimum (Min)** – Smallest value excluding outliers
- 2.**First Quartile (Q1)** – 25th percentile, lower edge of the box
- 3.**Median (Q2)** – 50th percentile, line inside the box
- 4.**Third Quartile (Q3)** – 75th percentile, upper edge of the box
- 5.**Maximum (Max)** – Largest value excluding outliers

Optional:

- Outliers** are plotted individually beyond the whiskers.
- Interquartile Range (IQR) = $Q3 - Q1$** helps identify outliers (values $< Q1 - 1.5 \times IQR$ or $> Q3 + 1.5 \times IQR$).

How It Looks

- **Box:** Q1 to Q3
- **Line inside box:** Median
- **Whiskers:** Min to Max (excluding outliers)
- **Dots:** Outliers



Why Box Plots Are Useful

- Summarizes **distribution and spread** in one visual.
- Shows **median and quartiles** clearly.
- Detects **outliers easily**.
- Useful for **comparing multiple datasets** side by side.

Quick Tip

Histogram shows the **shape of the distribution**, while a Box Plot shows **spread, central value, and outliers**.

3. Density Plot

A **Density Plot** is a **smoothed version of a histogram** that shows the **probability distribution of a continuous variable**.

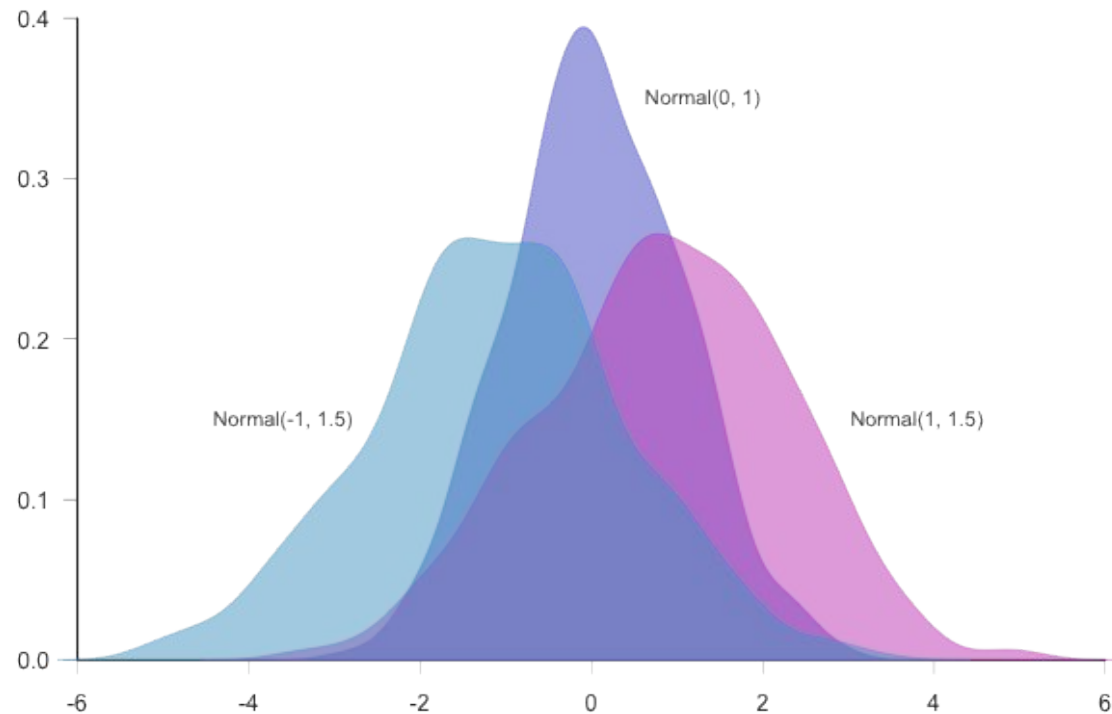
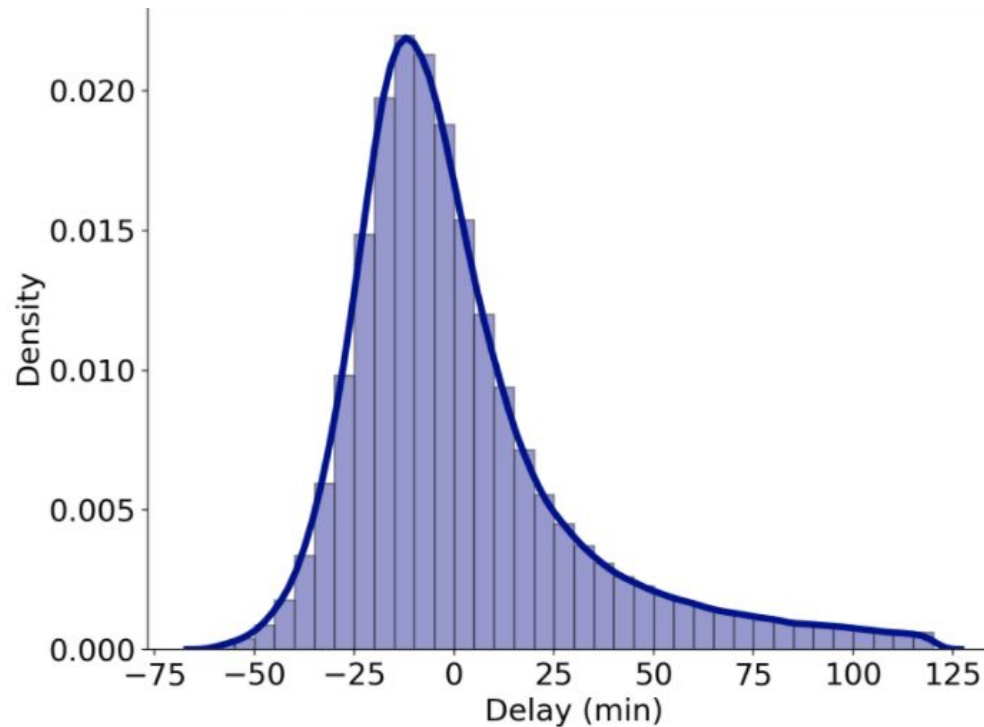
- It estimates the **distribution of data points** across the range.
- The **area under the curve equals 1**.

Key Features

- 1.Continuous Curve:** Unlike histograms with bars, density plots use a smooth line.
- 2.Peak:** Shows where data values are concentrated.
- 3.Spread:** Width of the curve shows variability.
- 4.Skewness:** Curve shape indicates asymmetry.
- 5.KDE (Kernel Density Estimation):** Most density plots use KDE to smooth the data.

Purpose

- Visualize the **distribution of a continuous variable**.
- Compare **multiple distributions** on the same plot.
- Identify **modes, skewness, and spread** more clearly than histograms.
- Useful in **statistical modeling and exploratory data analysis (EDA)**.



Multivariable Summaries

Multivariable Summaries describe and summarize **relationships between two or more variables** in a dataset.

- Unlike **single-variable summaries**, which focus on **one variable**, multivariable summaries help understand **interactions, correlations, and patterns** between variables.
- Useful in **exploratory data analysis (EDA)** and **predictive modeling**.

Types of Multivariable Summaries

1. Numerical vs Numerical Variables
2. Numerical vs Categorical Variables
3. Categorical vs Categorical Variables
4. High-Dimensional Summaries

A. Numerical vs Numerical Variables

- **Covariance:** Measures how two variables change together.

- Positive → both increase together
- Negative → one increases while the other decreases
- Can take **any value** ($-\infty$ to $+\infty$).

In short,

Covariance → tells **whether** variables move together.

Correlation → tells **how strongly** they move together.

- **Correlation Coefficient (Pearson, Spearman):** Measures **strength and direction** of linear (Pearson) or monotonic (Spearman) relationship. Values range: -1 to +1

- **Scatter Plot:** Graphical summary to visualize relationship. Example: Height vs Weight

- **Pair Plot / Scatterplot Matrix:** Shows pairwise relationships among several numerical variables.

Correlation

Correlation measures the **strength and direction of a relationship** between **two variables**.

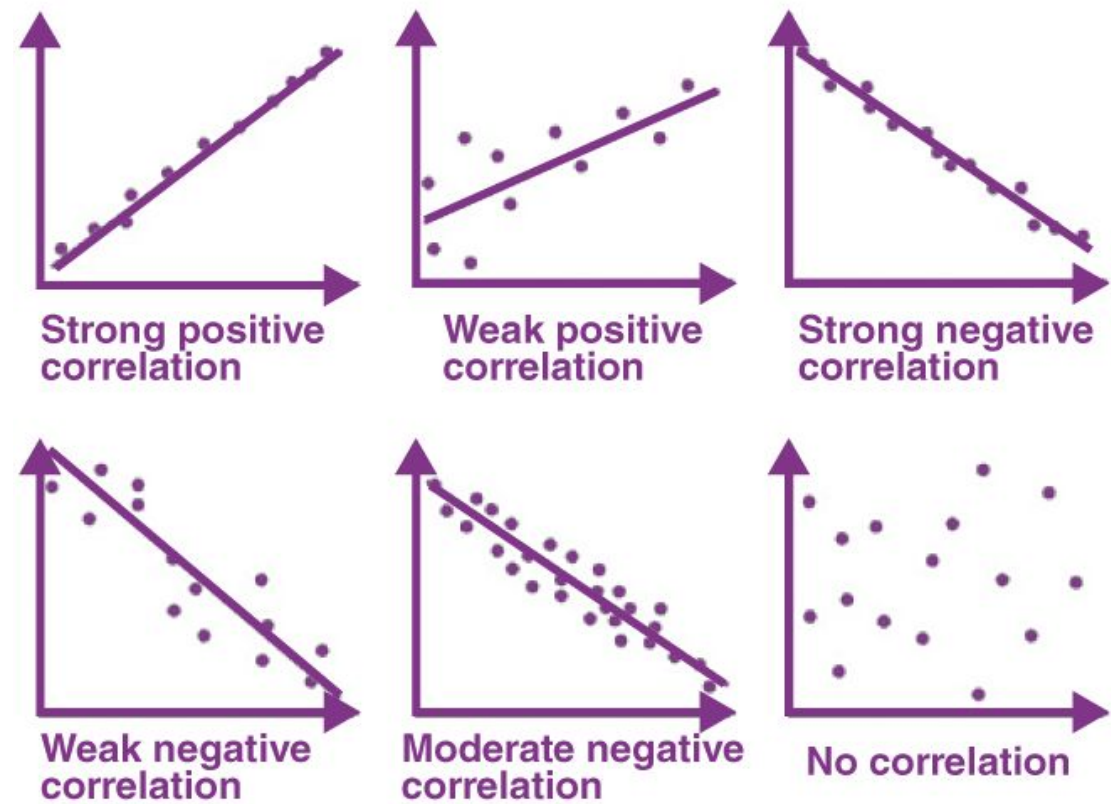
- It tells us **how one variable changes as another changes**.
- The most common type is **Pearson correlation**, which measures **linear relationships**.

Key Properties:

- Range: -1 to +1
 - **+1** → perfect positive correlation (both increase together)
 - **-1** → perfect negative correlation (one increases, the other decreases)
 - **0** → no linear correlation

Example:

- Height and Weight of individuals → positive correlation (~0.8)
- Temperature and Ice Cream Sales → positive correlation



Spurious Correlation

A **spurious correlation** occurs when **two variables appear correlated**, but **there is no direct causal relationship**.

- Often caused by **coincidence, or confounding variables**.
- Misleading because it **suggests a relationship that doesn't exist causally**.

Example:

- Ice cream sales and drowning incidents → correlated, but **both increase in summer** (temperature is the lurking variable).
- Number of storks and human birth rate in some regions → coincidental correlation.

Key Point:

Correlation $\neq$ Causation

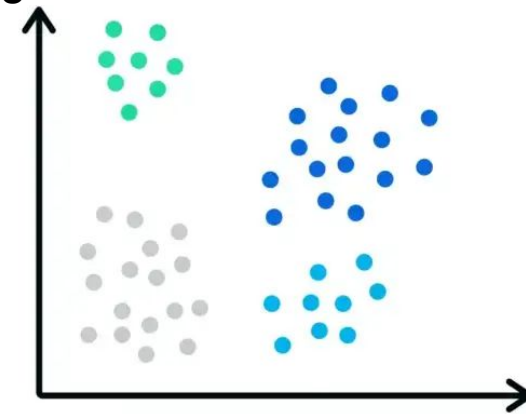
B. Numerical vs Categorical Variables

- **Grouped Summary Statistics:** Compare numerical data across categories
 - Mean, median, SD by category
 - Example: Average salary by department
- **Box Plot / Violin Plot:** Show distribution of numerical variable across categories
- **Bar Chart / Clustered Bar Chart:** Show summary statistics across groups.

	Dog	Cat	Total
Male	42	10	52
Female	9	39	48
Total	51	49	100

C. Categorical vs Categorical Variables

- **Contingency Table (Cross Tab):** Shows frequency counts for combinations of categories
 - Example: Gender vs Product Preference
- **Stacked / Grouped Bar Charts:** Graphical summary of relationships.



D. High-Dimensional Summaries

- **Correlation Matrix / Heatmap:** Shows pairwise correlations among many numerical variables.
- **Principal Component Analysis (PCA):** Summarizes multivariate numerical data into fewer dimensions.
- **Cluster Analysis:** Groups similar observations based on multiple variables.

Example

Suppose we have a dataset with **Height, Weight, Gender**:

- 1.Height vs Weight (Numerical/Numerical):** Scatter plot, correlation = 0.85 → strong positive relationship
- 2.Weight by Gender (Numerical/Categorical):** Box plot → compare median and spread
- 3.Gender vs Diet Preference (Categorical/Categorical):** Contingency table → counts of each combination

Key Points

- Multivariable summaries **reveal relationships** that single-variable summaries cannot.
- Helps in **feature selection, detecting interactions, and exploratory analysis**.
- Both **numerical and graphical methods** are important for interpretation.